☁️ Cloud Storage Cost Analysis Report

Resume Storage Solution Comparison

Generated: November 5, 2025 | Exchange Rate: 1 USD = ₹83.5

📊 **Executive Summary:** This comprehensive report analyzes cloud storage costs for resume storage, comparing AWS S3, Azure Blob Storage, and SQL Server solutions. Based on 80 resumes uploaded daily at 100 KB each.

1. Input Parameters

80
Resumes per Day

100 KB
Average Resume Size

29,200
Resumes per Year

2.78 GB
Storage per Year

2. AWS S3 Cost Projections

2.1 Short-term Costs

Period	Resumes	Storage	Storage Cost	PUT Cost	Total (INR)
1 Day	80	0.0076 GB	₹0.01	₹0.03	**₹0.05**
1 Month	2,400	0.23 GB	₹0.44	₹1.00	**₹1.44**

2.2 Long-term Costs (1-5 Years)

Period	Total Resumes	Storage (GB)	Storage Cost	PUT Cost	Total Cost
1 Year	29,200	2.78 GB	₹5.01	₹12.53	**₹17.54**

2 Years	58,400	5.57 GB	₹10.86	₹24.21	**₹35.07**
3 Years	87,600	8.35 GB	₹15.87	₹36.74	**₹52.60**
5 Years	146,000	13.92 GB	₹26.72	₹60.95	**₹87.67**

💡 **Amazing Value:** Store 146,000 resumes for 5 years for less than ₹100!

3. Storage Solution Comparison

3.1 Pricing Comparison (10 GB/month)

Storage Solution	Monthly Cost (USD)	Monthly Cost (INR)	Difference	Best For
Azure Blob Storage	$0.18	₹15	✅ Cheapest	Microsoft ecosystem
AWS S3	$0.23	₹19	+27%	AWS ecosystem
SQL Server / Azure SQL	$1.15	₹96	+505%	NOT for files ❌

⚠️ **Critical Warning:** SQL Server is 5-6x more expensive and NOT designed for file storage. Use object storage (S3 or Blob) instead!

3.2 Feature Comparison

Feature	AWS S3	Azure Blob	SQL Server
Type	Object Storage	Object Storage	Database
Scalability	Unlimited ♾️	Unlimited ♾️	Limited 📊
Best Use	Files, backups, media	Files, backups, media	Structured data only
Performance	High throughput	High throughput	Query optimization

4. AWS S3 Storage Classes

Storage Class	INR/GB/month	Savings vs Standard	Best Use Case
S3 Standard	₹1.92	-	Frequently accessed (0-30 days)
S3 Standard-IA	₹1.04	46% cheaper	Infrequent access (30-90 days)
S3 One Zone-IA	₹0.83	57% cheaper	Non-critical data
S3 Glacier Instant	₹0.33	83% cheaper	Archive with instant access
S3 Glacier Flexible	₹0.30	84% cheaper	Archive (90+ days)
S3 Glacier Deep Archive	**₹0.08**	**96% cheaper**	**Long-term archive (1+ year)**

5. Cost Optimization Strategies

🎯 Strategy #1: Use S3 Intelligent-Tiering

- Automatically moves data to optimal storage tier
- **Save up to 68%** on storage costs
- No retrieval fees or operational overhead
- Perfect for unpredictable access patterns

🎯 Strategy #2: Implement Lifecycle Policies

- **0-30 days:** S3 Standard (fast access)
- **30-90 days:** S3 Standard-IA (save 46%)
- **90+ days:** S3 Glacier (save 83%)
- **1+ year:** Deep Archive (save 96%)

Result: 5-year cost drops from ₹87.67 to approximately ₹15-20!

🎯 Strategy #3: Compress Files (50-70% reduction)

- PDF compression: 100 KB → 30-50 KB
- Use libraries: pdf-lib, Ghostscript, or pdfcpu
- Direct 50% savings on storage costs
- Faster uploads and downloads

🎯 Strategy #4: Implement Caching

- Cache frequently accessed resumes in Redis/Memory
- Use CloudFront CDN for global delivery
- Reduce GET request costs by 80-90%
- Faster response times for users

🎯 Strategy #5: AWS Free Tier (First 12 months)

- ✅ 5 GB storage (covers your first year!)
- ✅ 20,000 GET requests/month
- ✅ 2,000 PUT requests/month (partial coverage)
- ✅ 15 GB data transfer out

6. Projected Savings with Optimization

Strategy	5-Year Cost	Savings
S3 Standard (No Optimization)	₹87.67	Baseline
With 50% Compression	₹43.84	-50%
With Lifecycle to Glacier (90 days)	₹26.30	-70%
Compression + Lifecycle	**₹13.15**	**-85%**

🏆 Best Case Scenario

5-Year Total Cost: ₹13-15 (with all optimizations)
That's less than ₹3 per year for storing 29,200 resumes annually!

7. Final Recommendation

✅ Recommended Solution: AWS S3 with Optimization

Why AWS S3:

- ✅ Extremely cost-effective (₹13-20 for 5 years)
- ✅ Unlimited scalability
- ✅ 99.999999999% durability (11 nines)
- ✅ Robust security (encryption, access control)
- ✅ Easy integration with other services
- ✅ Mature ecosystem and documentation

Implementation Steps:

1. Create AWS S3 bucket with versioning enabled
2. Set up lifecycle policy (Standard → IA → Glacier)
3. Implement file compression before upload
4. Configure CloudFront for CDN delivery
5. Set up monitoring with AWS CloudWatch

🔄 Alternative: Azure Blob Storage

Choose Azure Blob if:

- You're already using Microsoft Azure ecosystem
- You want 15% lower costs (₹15 vs ₹19 per 10GB)

- You need Azure AD integration

8. Security & Compliance

Security Feature	AWS S3	Azure Blob
Encryption at Rest	✅ AES-256, KMS	✅ AES-256, SSE
Encryption in Transit	✅ TLS/SSL	✅ TLS/SSL
Access Control	✅ IAM, Bucket Policies	✅ Azure AD, SAS
Compliance	✅ ISO, SOC, HIPAA, GDPR	✅ ISO, SOC, HIPAA, GDPR
Versioning	✅ Built-in	✅ Snapshots
Backup	✅ Cross-region replication	✅ Geo-redundancy

9. Important Notes

- 📍 Prices are for US East (N. Virginia) region
- 🔄 Costs vary by AWS region (typically ±10%)
- 📊 GET request costs not included (minimal for typical usage)
- 🌐 Data transfer costs apply after 100 GB/month
- ⚠️ Always verify current pricing on official AWS website
- 🧪 Test with pilot program before full deployment
- 📈 Monitor costs with AWS Cost Explorer

10. Useful Resources

- **AWS S3 Pricing:** https://aws.amazon.com/s3/pricing/
- **AWS Pricing Calculator:** https://calculator.aws
- **AWS S3 Documentation:** https://docs.aws.amazon.com/s3/
- **Azure Blob Pricing:** https://azure.microsoft.com/pricing/details/storage/blobs/

- **AWS Free Tier:** https://aws.amazon.com/free/
